Andhra University College of Engineering

Electronics and Communication Engineering | Course Code: EC301

Digital Signal Processing - Digital Signal Processing Transform & Filter De

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Overview: Radix-2 decimation-in-time and decimation-in-frequency FFT butterflies, Butterworth &

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Discrete Fourier Transform & FFT
Properties of DFT, Circular convolution, Radix-2 DIT-FFT and DIF-FFT algorithms

Unit 2: IIR Filter Design
Bilinear transformation method, Impulse invariance method, Analog prototype transformations

Unit 3: FIR Filter Design
Symmetric and antisymmetric FIR filters, Windowing techniques Rectangular, Hamming, Hanning, Blackma

Unit 4: Finite Word Length Effects
Coefficient quantization, Round-off noise in digital filters, Limit cycle oscillations

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Compute the 8-point DFT using Radix-2 Decimation in Time DIT FFT butterfly computation.
2. Design an analog Butterworth low-pass filter satisfying specified passband and stopband attenuations.
3. Formulate the transfer function of a linear phase FIR high-pass filter using the Hamming window techni

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* Alan V. Oppenheim, Ronald W. Schafer - Discrete-Time Signal Processing Pearson
* John G. Proakis, Dimitris G. Manolakis - Digital Signal Processing Pearson 4th Ed

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